Homework3Notebook

February 17, 2026

1 Homework 3

1.1 2)

For a

[1]:
```
#constants
me = 0.000511
mmu = 0.1057
mp = 0.9383

beam_energies = [0.5, 1.0, 2.0, 10.0, 100.0]
```

[2]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

[3]:
```
def calc_T_max(E, mass):
    gamma = E/mass
    print(E, mass)
    beta = np.sqrt(1-1/gamma**2)

    return (2.0 * me * (beta*gamma)**2)/(1+(2*gamma*me/mass +(me/mass)**2))
```

[4]:
```
def calculate_angle(beam_energies, mass):
    T_results = [[None]*1000]*len(beam_energies)
    angle_results = [[None]*1000]*len(beam_energies)

    for E, energy in enumerate(beam_energies):

        T_max = calc_T_max(energy, mass)

        T_vals = np.linspace(1e-12, 0.999*T_max, 1000)

        thetas = np.arccos(np.sqrt(T_vals/T_max))

        angle_results[E] = thetas
        T_results[E] = T_vals
```

```python
    return T_results, angle_results
```

[5]:
```python
mu_Ts, mu_angles = calculate_angle(beam_energies, mmu)
```

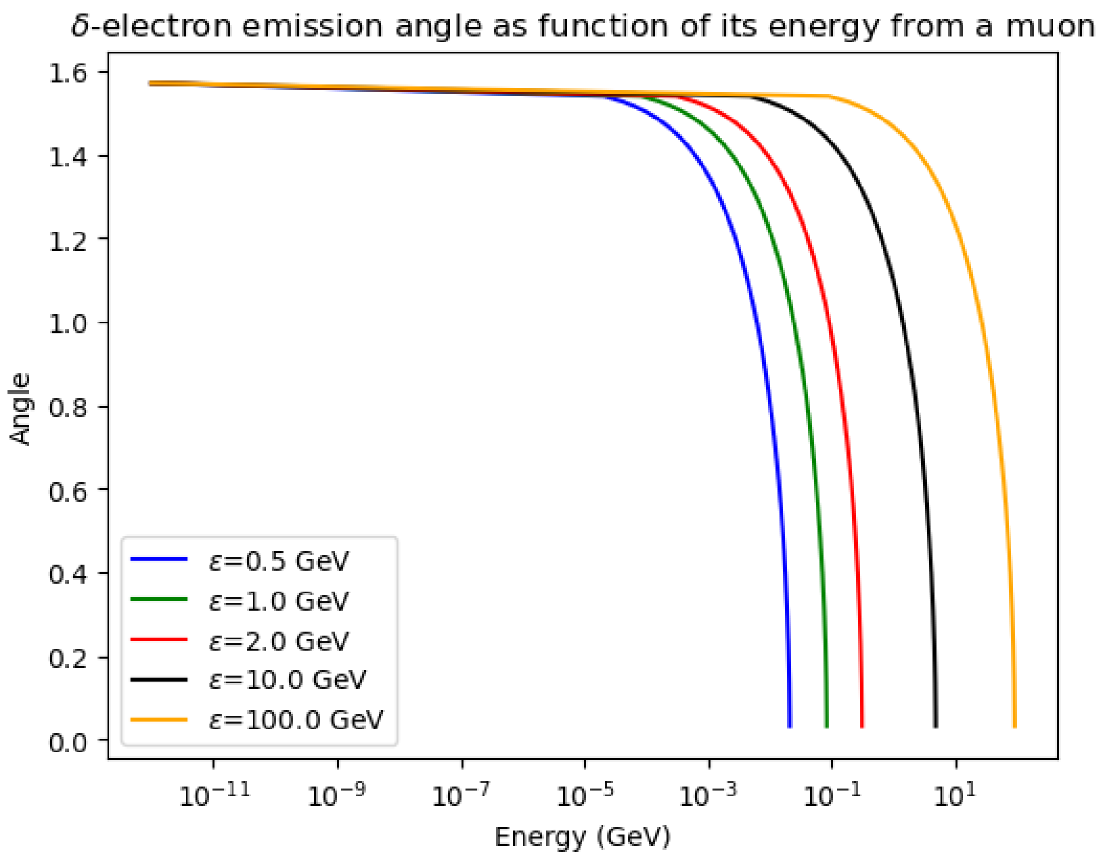
```
0.5 0.1057
1.0 0.1057
2.0 0.1057
10.0 0.1057
100.0 0.1057
```

[6]:
```python
fig, ax = plt.subplots()
facecolors = ["Blue", "Green", "Red", "Black", "Orange"]

for i in range(len(facecolors)):
    ax.plot(mu_Ts[i], mu_angles[i], color=facecolors[i],␣
 ↪label=f"$\epsilon$={beam_energies[i]} GeV")
    ax.set_xscale("log")

ax.legend()
ax.set_xlabel("Energy (GeV)")
ax.set_ylabel("Angle")
ax.set_title(r"$\delta$-electron emission angle as function of its energy from␣
 ↪a muon")
```

[6]:
```
Text(0.5, 1.0, '$\\delta$-electron emission angle as function of its energy from
a muon')
```



[7]:
```
p_Ts, p_angles = calculate_angle(beam_energies, mp)
```

```
0.5 0.9383
1.0 0.9383
2.0 0.9383
10.0 0.9383
100.0 0.9383
```

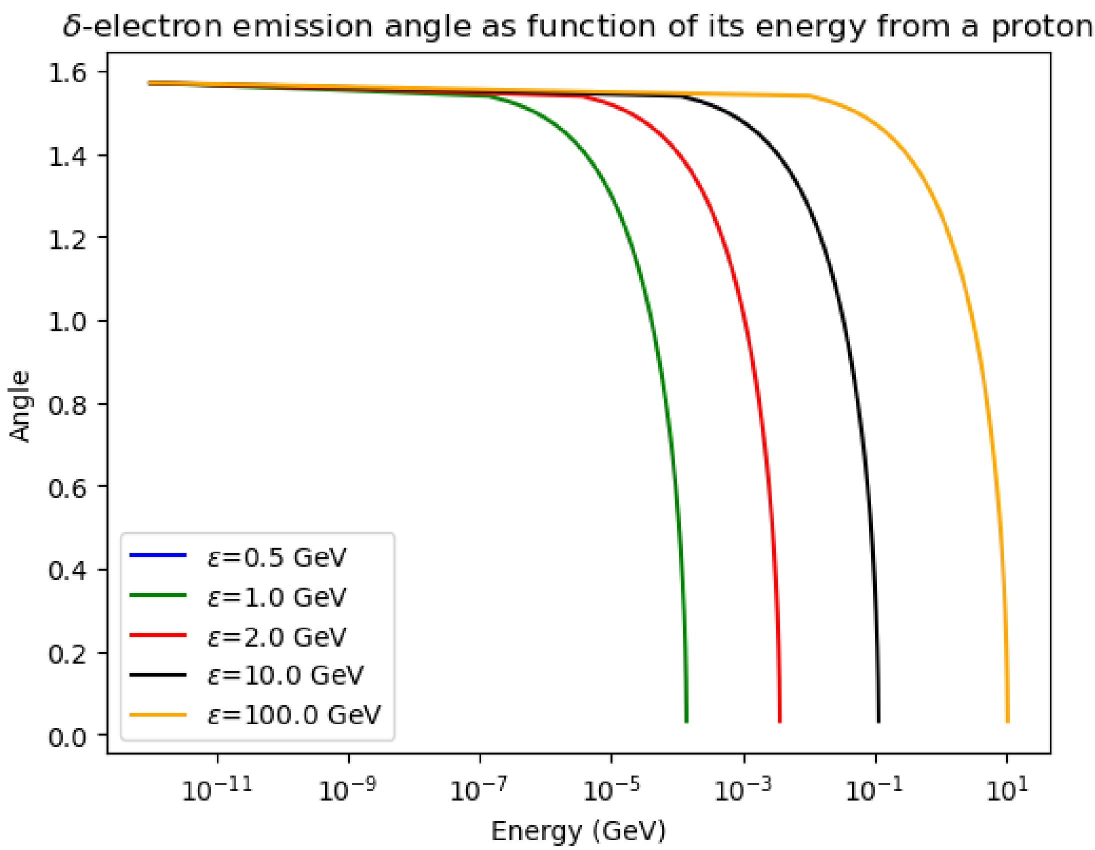
```
/tmp/ipykernel_37046/117918647.py:4: RuntimeWarning: invalid value encountered
in sqrt
  beta = np.sqrt(1-1/gamma**2)
```

[8]:
```
fig, ax = plt.subplots()
facecolors = ["Blue", "Green", "Red", "Black", "Orange"]

for i in range(len(facecolors)):
    ax.plot(p_Ts[i], p_angles[i], color=facecolors[i],␣
 ↪label=f"$\epsilon$={beam_energies[i]} GeV")
    ax.set_xscale("log")
```

```python
ax.legend()
ax.set_xlabel("Energy (GeV)")
ax.set_ylabel("Angle")
ax.set_title(r"$\delta$-electron emission angle as function of its energy from a proton")
```

[8]:
```
Text(0.5, 1.0, '$\\delta$-electron emission angle as function of its energy from
a proton')
```



This implies that the majority of δ-electrons will be produced in the forwards direction, especially the ones that contribute most to energy loss. These will contribute to the development of things like electromagnetic showers. In tracking detectors, this will mean that hits with larger energy deposits may be physically larger, and in calorimeters, will appear as the classic EM shower shape.